Connecting & Testing Peripherals Prac.

1. Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

Connect the Webcam

- Plug the USB cable of your webcam into an available USB port on your computer.
- Most modern webcams are **plug-and-play** and do not require manual driver installations. Windows automatically detects the device and installs the generic USB Video Device driver.

Verify and Install/Update Drivers (If Needed)

- Press the **Windows Key + S**, type Device Manager, and open it.
- Expand the **Cameras** or **Imaging devices** category.
- If your webcam appears with a yellow warning icon or is missing, right-click the device and select **Update driver** -> **Search automatically for drivers**.
- Alternatively, if your webcam is a specialized or high-end model with custom features, download the official software or driver package directly from the manufacturer's website (such as Logitech or Dell).

Check Privacy Permissions

- Open **Settings** from the Start menu.
- Go to **Privacy & security** > **Camera**.
- Ensure that **Camera access** and **Let apps access your camera** are toggled **On**.

Test the Webcam and Capture a Photo

- Press the **Windows Key + S**, type Camera, and open the **Camera** app.
- If your computer has a built-in camera and the wrong feed is showing, click the **switch camera button**(picture flipper) in the top-right corner of the app window to select your USB webcam.
- Click the on-screen **shutter/camera button** to capture a photo.
- Your newly captured photo will appear in the **Camera Roll** thumbnail preview in the bottom-right corner.

2. Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Step 1: Connect the Printer

- **For a USB Printer:**

    1. Turn on the printer.
    2. Plug the USB cable directly into an available USB port on your computer.
    3. Wait a few moments for Windows to recognize the device and attempt an automatic setup.

- **For a Network Printer (Wi-Fi or Ethernet):**
    1. Connect the printer to your router or local network via Wi-Fi or an Ethernet cable.
    2. Make sure your computer is connected to the **same network** as the printer.
    3. Keep the printer's IP address handy (you can find this on the printer's control panel or network configuration sheet).

Step 2: Install the Driver

Windows will often install a generic driver automatically, but for full functionality, manufacturer-specific drivers are best:

1. Open Start > Settings > Bluetooth & devices > Printers & scanners.
2. Click **Add device** (or Add a printer or scanner).

    - If your printer appears in the list, click **Add device** next to it.
    - **For a network printer using an IP:** Click *Add manually* or *"The printer that I want isn't listed,"*then select **Add a printer using a TCP/IP address or hostname** and enter your printer's IP address.

3. If Windows cannot find or install the right software, go to the official support page of your printer manufacturer (such as HP, Canon, Epson, or Brother) to download the latest driver package for your OS version.
4. Run the downloaded installer file (or point the manual installer to the extracted .INF file if installing via Device Manager/Add Printer wizard) and follow the on-screen prompts.

Step 3: Print a Test Page

1. Go back to **Start** > **Settings** > **Bluetooth & devices** > **Printers & scanners** (or open the classic **Control Panel** > **Devices and Printers**).
2. Click on your newly installed printer.
3. Select **Printer properties** or **Manage device**.
4. Click **Print a test page** to confirm that communication between your computer and the printer is working properly.

3. Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

To simulate and resolve a printer connectivity issue, I would follow these steps:

1. Disconnect the printer: First, I would unplug the printer's USB cable from the computer while keeping the printer switched on.

2. Open Device Manager: Press `Windows + X` and select Device Manager from the menu.

3. Check for errors: I would look under Printers, Universal Serial Bus controllers, or Other devices to check whether the printer is missing or showing a yellow warning icon.

4. Reconnect the printer: I would plug the USB cable back into the computer and ensure that it is properly connected.

5. Scan for hardware changes: In Device Manager, I would click the Action menu and select Scan for hardware changes. Windows will search for the reconnected printer and attempt to detect it.

6. Verify the connection: Once the printer appears in Device Manager without any warning icons, I would open Settings → Bluetooth & devices → Printers & scanners to check whether the printer is detected.

Conclusion: By reconnecting the USB cable and using the Scan for hardware changes option, I can help Windows detect the printer again. This resolves the issue if it was caused by a loose or disconnected USB connection.

4. A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

If the webcam works in the default Camera app but does not display video in Zoom, I would follow these steps:

1. Close other applications: I would close the default Camera app and any other applications that might be using the webcam, as some applications may prevent Zoom from accessing it.

2. Check Zoom camera settings: I would open Zoom, go to Settings → Video & effects and check whether the correct webcam is selected under the Camera option.

3. Check camera permissions: I would open Windows Settings → Privacy & security → Camera and ensure that:

- Camera access is turned ON.
- Let apps access your camera is enabled.
- Let desktop apps access your camera is enabled.

4. Restart Zoom: I would completely close Zoom and reopen it to refresh the webcam connection.

5. Test the webcam: I would start a Zoom test meeting and check whether the video is displaying correctly.

6. Update Zoom and webcam drivers: If the problem continues, I would update Zoom to its latest version and check Device Manager for webcam driver updates.

Conclusion: The webcam was working properly in the Camera app, so the issue was likely related to Zoom's camera selection, permissions, or another application using the webcam. After checking these settings and restarting Zoom, I would test the webcam again to confirm that it works in both applications.